Statement of Research Interests

Paolo Astrino

My professional and academic journey is defined by a transition from the quantitative foundations of economics to the applied frontiers of artificial intelligence. This path has been driven by a singular objective: to build practical, scalable, and interpretable systems that solve real-world complexities through machine learning and automation.

My research interests primarily lie at the intersection of **Machine Learning**, **Automation**, and **Economics**. This unique combination allows me to approach AI not just as a technical challenge, but as a socio-economic tool that requires efficiency and transparency to be truly impactful. My recent work, including arXiv publications, explores the intersection between advanced analytics and applied AI, with a specific emphasis on Retrieval-Augmented Generation (RAG) and privacy-first local processing.

The cornerstone of my recent research is the **CUBO** project (arXiv:2602.03731), where I engineered a self-contained RAG system optimized for consumer hardware. This project required resolving critical trade-offs between hybrid retrieval models and O(1) memory scaling—ensuring that powerful AI capabilities remain accessible in privacy-critical environments without the need for cloud dependency. My previous work on **Local Hybrid Retrieval-Augmented Document QA** (arXiv:2511.10297) further established this theme, bridging the gap between semantic understanding and keyword precision to create robust, autonomous machine learning pipelines.

As a "Builder-Researcher," I prioritize the end-to-end lifecycle of a solution. My background as a Project Manager and Full Stack Developer informs my research by grounding it in industrial-grade feasibility. I believe that the most impactful research is that which identifies a root business or societal bottleneck and engineers a high-impact solution to eliminate it.

Moving forward, I intend to continue exploring how automated intelligence can be integrated into existing frameworks to enhance decision-making speed and accuracy, particularly in contexts where data sovereignty and operational efficiency are paramount. I am committed to developing AI-native workflows that prioritize user autonomy and high-utility outcomes.

Paolo Astrino
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Portfolio: https://paoloastrino.github.io/